WORKED SOLUTION

STEP II 1999, Question 9

Mechanics • Variable Force Motion

STEP II 1999, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q9

The key idea

The acceleration depends only on the distance x, not on time, so the right tool is $\ddot{x} = v\,\dfrac{\mathrm{d}v}{\mathrm{d}x}$. This turns the equation of motion into one we can integrate directly to get v^2 as a function of x (an energy equation in disguise). Everything else is about asking where v^2 hits zero: that is where a fragment stops and turns back.

Solution

We model each fragment as a particle moving along a straight line through the fixed point G, with x its distance from G and $v = \dot{x}$ its velocity, positive when moving away from G. Initially $x = 0$ and $v = u > 0$. Because the acceleration points towards G, it is negative in this convention.

Part (i)

For $x < R$ we have $\ddot{x} = -k_1 x^3$, and writing $\ddot{x} = v\,\dfrac{\mathrm{d}v}{\mathrm{d}x}$,

$$v\frac{\mathrm{d}v}{\mathrm{d}x} = -k_1 x^3.$$

Separating and integrating from the start ($x = 0$, $v = u$):

$$\tfrac{1}{2}v^2 - \tfrac{1}{2}u^2 = -\tfrac{1}{4}k_1 x^4, \qquad \text{so} \qquad v^2 = u^2 - \tfrac{1}{2}k_1 x^4 \quad (x < R).$$

While the fragment is still moving outwards, $v = \sqrt{u^2 - \tfrac{1}{2}k_1 x^4}$.

Part (ii)

With $2a^2 = k_1R^4$, the formula from (i) reads $v^2 = u^2 - a^2(x/R)^4$. As x increases from 0, v^2 decreases, and it reaches zero at

$$x_0 = \left(\frac{2u^2}{k_1}\right)^{1/4}.$$

If $u < a$ then $2u^2 < 2a^2 = k_1R^4$, so $x_0 < R$.

Ask yourself The speed is zero at x_0, but why does that mean the fragment never gets further?

At x_0 the fragment is momentarily at rest, but its acceleration there is $-k_1x_0^3 \neq 0$, directed towards G. So it is not in equilibrium: it starts to move back towards G. In fact x can never exceed x_0 at all, because the formula from (i) would then give $v^2 < 0$. The furthest the fragment ever gets is $x_0 < R$, so it never reaches a distance R from G, as required.

Part (iii)

First, a fragment with $u \geq b$ certainly reaches R. From the definitions,

$$b^2 = \tfrac{1}{2}k_1R^4 + \frac{2k_2}{3R^3} = a^2 + \frac{2k_2}{3R^3} > a^2,$$

so for $x \leq R$, part (i) gives

$$v^2 = u^2 - \tfrac{1}{2}k_1x^4 \geq u^2 - a^2 \geq b^2 - a^2 = \frac{2k_2}{3R^3} > 0.$$

In particular the fragment arrives at $x = R$ with $v_R^2 = u^2 - \tfrac{1}{2}k_1R^4 = u^2 - a^2$.

For $x \geq R$ the equation of motion is $v\dfrac{\mathrm{d}v}{\mathrm{d}x} = -k_2x^{-4}$. Integrating from $x = R$:

$$\tfrac{1}{2}v^2 - \tfrac{1}{2}v_R^2 = \frac{k_2}{3}\left(\frac{1}{x^3} - \frac{1}{R^3}\right),$$

so

$$\begin{aligned} v^2 &= u^2 - \tfrac{1}{2}k_1R^4 - \frac{2k_2}{3R^3} + \frac{2k_2}{3x^3} \\ &= u^2 - b^2 + \frac{2k_2}{3x^3} \qquad (x \geq R). \end{aligned}$$

That combination is exactly why b was defined as it is: b^2 is the total "cost" in $\frac{1}{2}v^2$ of escaping from G to infinity.

If $u \geq b$ then $u^2 - b^2 \geq 0$ and $\frac{2k_2}{3x^3} > 0$, so $v^2 > 0$ for every $x \geq R$ (this includes the borderline case $u = b$). Together with the bound for $x \leq R$, v^2 is strictly positive at every distance the fragment could reach.

The velocity v is continuous in time and starts at $u > 0$. To become negative it would have to pass through 0, which would need $v^2 = 0$ at some position, and we have just shown that never happens. So $v > 0$ throughout: from the moment of the explosion the fragment is always moving away from G. (Since $v^2 \geq 2k_2/(3x^3)$, it keeps going, with $x \to \infty$.)

Part (iv)

Now $a < u < b$. Since $u > a$, the argument of (iii) shows that $v^2 \geq u^2 - a^2 > 0$ on $[0, R]$, so the fragment reaches R still moving outwards. Beyond R,

$$v^2 = \frac{2k_2}{3x^3} - (b^2 - u^2),$$

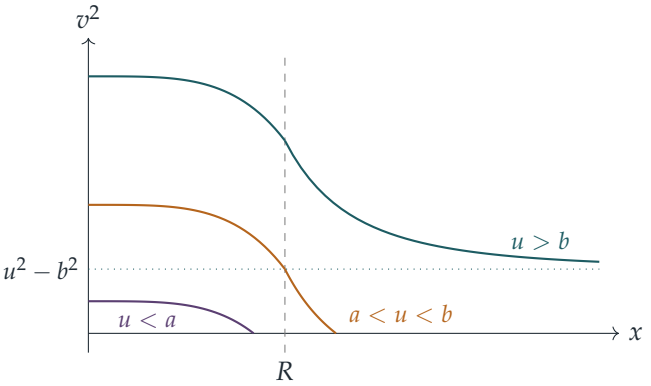
and now $b^2 - u^2 > 0$. This decreases as x increases, and is zero when

$$x^3 = \frac{2k_2}{3(b^2 - u^2)}.$$

This point does lie in the region $x \geq R$: since $u > a$, $b^2 - u^2 < b^2 - a^2 = \frac{2k_2}{3R^3}$, so $x^3 > R^3$. As in (ii), the fragment stops there with a non-zero acceleration $-k_2 x^{-4}$ towards G, and it cannot go further because v^2 would be negative. The maximum distance from G is

$$\boxed{x_{\max} = \left(\frac{2k_2}{3(b^2 - u^2)}\right)^{1/3}}$$

The figure shows v^2 against x for $k_1 = 2$, $k_2 = 3$, $R = 1$ (so $a^2 = 1$, $b^2 = 3$) and the three kinds of initial speed. Each fragment travels until its curve meets the axis.



Checks

- With $k_1 = 2$, $k_2 = 3$, $R = 1$ we get $a = 1$, $b = \sqrt{3}$; for $u = \sqrt{2}$ the formula gives $x_{\max} = (2 \cdot 3/(3 \cdot 1))^{1/3} = 2^{1/3} \approx 1.26$, just beyond R, as it should be.

- As $u \to a^+$, $b^2 - u^2 \to 2k_2/(3R^3)$ and $x_{\max} \to R$; as $u \to b^-$, $x_{\max} \to \infty$. Both match parts (ii) and (iii) at the boundaries.
- Stepping the motion numerically (Runge–Kutta, twelve random choices of k_1, k_2, R) reproduces the formula for v^2, the turning point $(2u^2/k_1)^{1/4}$ when $u < a$, the value of $x_{\max}$ in (iv) to within 0.1%, and no reversal at all when $u \geq b$.

Common slips

- Integrating the second stage from $x = 0$ instead of from $x = R$ with $v_R^2 = u^2 - \frac{1}{2}k_1R^4$: the force law changes at R, so the constant changes too.
- Losing the sign: the acceleration is towards G, so $v\,\mathrm{d}v/\mathrm{d}x$ is *minus* k_1x^3 or k_2x^{-4}.
- In (iii), checking only $x \geq R$ and forgetting that v must also stay positive inside R.

Beyond STEP

The quantity $\frac{1}{2}v^2 + V(x)$, with $V'(x)$ equal to minus the acceleration, is conserved for any force depending only on position. Plotting V (a "potential well") tells you at a glance which motions are bounded and which escape, and b is just the escape speed for this well. The same reasoning gives escape velocity in real gravity, and is the starting point for the phase-plane analysis of nonlinear oscillators. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops energy diagrams like this in detail.

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